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Bit Design Optimization with AI-Enhanced Photometric Dull Grading

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Abstract

The continuous advancement of drilling technology has brought about the need for more sophisticated tools and methodologies to enhance the performance and efficiency of drilling operations. One such innovation is Cutter Analytics (CA) technology, which leverages a multi-platform approach to provide comprehensive insights into bit performance and wear. This article explores the five main platforms of CA technology: the digital acquisition system, AI-empowered analysis of digital inputs, 360-degree integrated view, big data analytics, and AI-driven optimization and recommendation capabilities.

Introduction

The drill bit cutting structure plays a pivotal role in determining overall drilling performance, directly influencing rate of penetration (ROP), bit durability, and operational efficiency. As such, accurate evaluation of the bit's dull condition—particularly the wear and damage on the cutting structure—has long been recognized as essential for improving bit design and optimizing future runs.

Historically, the process of capturing and storing dull bit information for design feedback has been manual, time-consuming, and inconsistent. Traditional dull grading methods require subjective visual inspection, physical tagging, manual data entry, and offline communication between field personnel and design teams. This often resulted in delays and incomplete datasets, limiting the designer's ability to make informed improvements.

Despite advancements in digital workflows, the challenge of systematically storing dull condition data along with the corresponding drilling parameters remains unresolved. In current field practices, most dull photographs are still taken using mobile phones by drillers, company representatives, or bit service personnel. These images are often stored locally, lack standardized metadata, and are not always integrated with contextual drilling data such as lithology, weight on bit (WOB), rotary speed (RPM), and downhole dynamics.

To address these limitations, this study explores an integrated approach that combines photometric dull grading, AI-generated dull coding statistics, and heat map visualization of wear zones. Furthermore, the data is incorporated into advanced bit design software, enabling iterative improvements and performance prediction. Most importantly, this article presents a real-world case study from an actual application, where

This paper aims to demonstrate how structured dull analysis and AI integration can lead to more intelligent design decisions and improved performance in demanding drilling environments.

The primary objective is to analyze bit cutting structure dull and wear using high-resolution photometric data, enabling a more objective and repeatable assessment compared to traditional manual grading techniques.

Figure 1 displays seven images illustrating different types of cutter failures, arranged in two rows. The top row shows four types of failures: WT (Worn Cutter), CT (Chipped Cutter), BT (Broken Cutter), and SP (Spalled Cutter). The bottom row shows three types of failures: DL (Delaminated), BF (Bond Failure), and LT (Lost Cutter). To the right of these images is a large image of a cutter head with colored labels (CT, BT, CT, CT, BF, P) indicating failure locations and a coordinate system (y1).

An integral part of this analysis involves the creation of damage zone heat maps. By mapping wear intensity and location on the bit face, these heat maps provide intuitive visual feedback, helping engineers to quickly identify high-stress regions and correlate wear with specific drilling parameters or formation characteristics.

The study also focuses on the integration of this enriched data into existing bit design software platforms. This integration enables designers to use real-world wear data to inform future bit designs, shorten development cycles, and improve performance predictions under similar operating conditions.

Ultimately, the goal is to develop actionable design recommendations based on a holistic analysis of dull grading, AI insights, and operational data. These recommendations aim to optimize bit geometry, material selection, and cutter placement for improved durability and efficiency in complex drilling scenarios.

Through this combined approach, the study contributes to a data-driven framework for continuous improvement in bit design, performance evaluation, and field application strategies.

Identify the Application Where Bit Design Faces Severe Cutting Structure Damages

The first step in the study was identifying a drilling application where bit failure due to cutting structure damage was particularly severe. (Fig.2). This was critical to ensure the effectiveness of any design enhancement or analytical method applied. For this case, a 16-inch polycrystalline diamond compact (PDC)

bit used in one of the most challenging drilling sections was selected. This section is characterized by extreme downhole conditions including hard interbedded formations, high levels of impact loading, and abrasive lithology. Bits deployed in this section frequently returned with significant damage to the cutters and matrix body, leading to reduced ROP, premature bit trips, and increased drilling costs. This made it a prime candidate for an advanced analytical study to understand dull mechanisms and to validate potential design improvements.

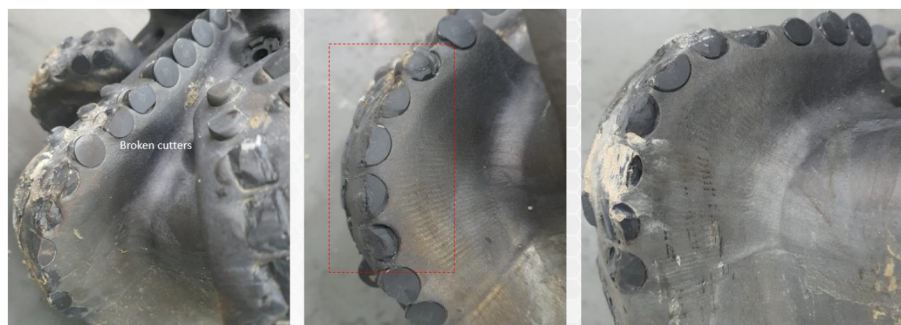


Figure 2—16-inch PDC bit dull condition.

Inspect the Bits Using Photometric Dull Grading AI Mobile Application

After retrieval from the wellbore, each dull bit was immediately subjected to a visual inspection using a photometric dull grading mobile application enhanced with artificial intelligence (AI). (Fig 3) The mobile application was developed to standardize dull grading and provide a consistent, objective view of the bit condition. The inspection process involved capturing high-resolution, multi-angle photographs of the bit under controlled lighting conditions. The AI component of the application provided real-time analysis and pre-tagged dull features, identifying damage such as chipping, breakage, crater wear, and thermal damage. This step marked a shift from subjective manual grading to a digitized and repeatable assessment, forming a solid foundation for further analytics.

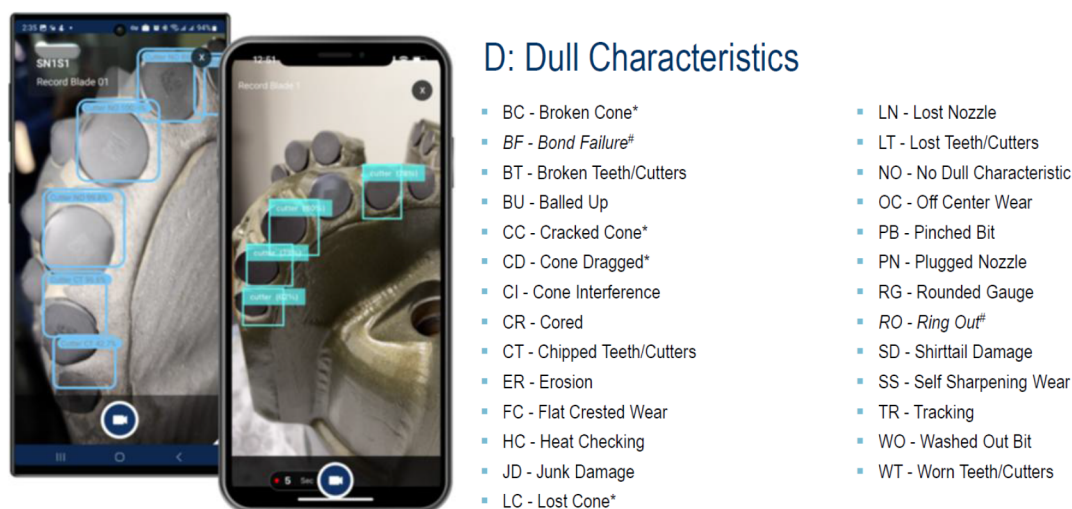


Figure 3—AI Mobile application for photo scanning bits

Uploading the Photos to the CA Platform

Once the photographs of the used bit were captured—they were uploaded to a cloud-based CA platform. This platform acts as both a centralized digital repository and a high-performance processing engine for

cutter dull grading and forensic analysis. The main objective of using such a platform is to standardize and automate the cutter evaluation process, eliminating subjectivity.

Upon upload, each dataset includes not only the images themselves but also a unique bit serial number, which acts as a digital key to unlock associated metadata from the cloud database. The serial number synchronization is a critical step in the analysis workflow. It enables the platform to automatically retrieve comprehensive information tied to that specific bit, including the bit type, design characteristics (blade count, cutter size), run history, formation data, and the complete drilling records such as depth drilled, ROP, torque, WOB and other operational parameters.

The platform leverages this contextual information in real time. Once synchronization is complete, it evaluates the quality of the incoming image set and immediately displays the percentage of cutters and blades successfully processed. This quality assurance metric provides a direct indication of the reliability of the analysis results. The higher the percentage of accurately detected cutters and blades, the more dependable the downstream interpretation will be. This validation step serves both as an internal check and a user-facing diagnostic tool, giving engineers a quick sense of confidence in the system's output (see Fig. 4).



Figure 4—Capturing and Uploading bits to CA platform

With image and metadata in hand, the platform then initiates its automated analytics engine. At the core of this engine are advanced machine learning (ML) models and computer vision algorithms, which have been trained on thousands of annotated cutter images from a wide range of field applications. These models are capable of recognizing a diverse array of wear patterns and failure modes with high accuracy and repeatability.

The first task the system performs is to locate and segment each cutter in the image set. Using image recognition and segmentation algorithms, it isolates individual cutters from the surrounding structure, even in cases of heavy wear, debris, or irregular lighting. This segmentation is critical, as it allows each cutter to be independently evaluated and assigned a condition score.

Once segmented, the system assesses each cutter's condition using a multi-parameter approach. It identifies dull codes according to IADC or custom-defined criteria, detects physical damage such as chipping, spalling, thermal cracking, and edge degradation, and quantifies wear level. Severity levels are automatically assigned using predefined thresholds based on cutter size, location (nose, shoulder, gauge), and functional role in the bit's design.

In addition to individual cutter metrics, the platform performs spatial analysis across the entire bit face. It maps cutter wear patterns across rows and blades to identify distribution trends, which can reveal design

inefficiencies or operational inconsistencies. For example, excessive wear concentrated on outer blades may indicate poor gauge protection or suboptimal hydraulic flow, while early failure on nose cutters might suggest overload conditions or poor weight transfer.

All the collected data—ranging from cutter-level metrics to bit-wide wear maps—is stored in a structured database for further review, reporting, and long-term analytics. Engineers can generate automated reports, compare current run data with historical performance, and track the evolution of cutter damage across multiple wells. This level of insight allows for continuous design improvement and operational optimization.

Understanding the Application Information with the Help of Data Integration Process

To provide meaningful context to the dull grading data, a comprehensive data integration process was employed. This involved linking bit images and condition data with drilling records for each run and serial number. (Fig 5.1) Through integration with the operator's drilling record system, the analytics platform was able to easily find the bit details from the existing database. If real-time data is available, the designer can easily pull the information regarding parameters such as WOB, RPM, torque, standpipe pressure, and vibration metrics. (Fig 5.2) Additionally, lithology logs and bottom-hole assembly (BHA) configurations were incorporated. This enabled benchmark analysis at the serial number level, allowing a detailed comparison of bit performance across multiple runs in the same formation type or hole section. This step was essential for understanding why certain cutters failed and identifying patterns tied to operational or geological conditions.

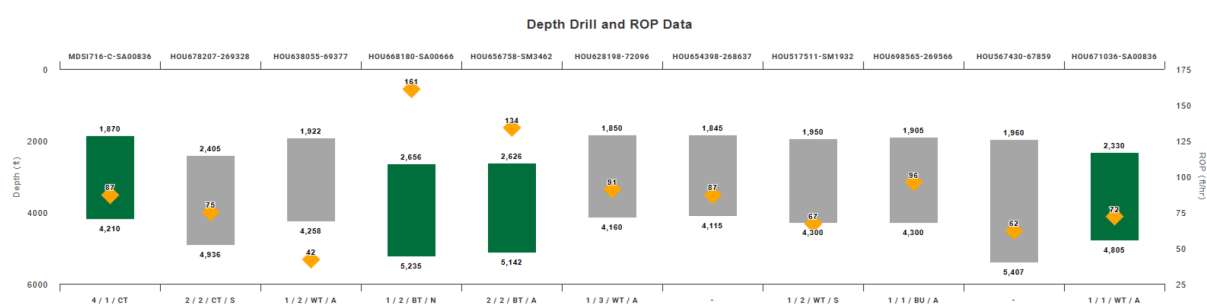


Figure 5.1—Bit actual performance from CA platform after uploading the bit serial number

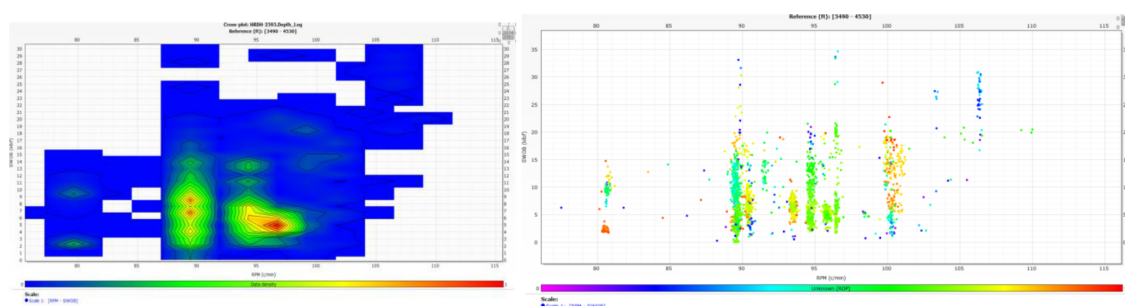


Figure 5.2—Bit actual performance from CA platform after uploading the bit serial number

AI-Generated Dull Grading Analysis

Following the data integration, the AI platform conducted a comprehensive dull grading analysis. This included cutter-by-cutter evaluation (Fig.6) and assignment of IADC dull and wear codes. (Fig.6.1) For each cutter, the platform provided a classification of wear types—such as mechanical abrasion, thermal damage, impact chipping, or delamination—based on a multi-parametric model that included visual characteristics and historical wear profiles. One of the key deliverables from this step was the generation of heat maps indicating the damage zones across the bit radius. (Fig.6.2) These heat maps visually highlighted areas with concentrated damage, providing an intuitive view of where stress and wear were most pronounced. This

cutter-level granularity helped identify systematic failure modes, such as edge cutter chipping or central cone damage, tied to the bit's profile and the operational loading.

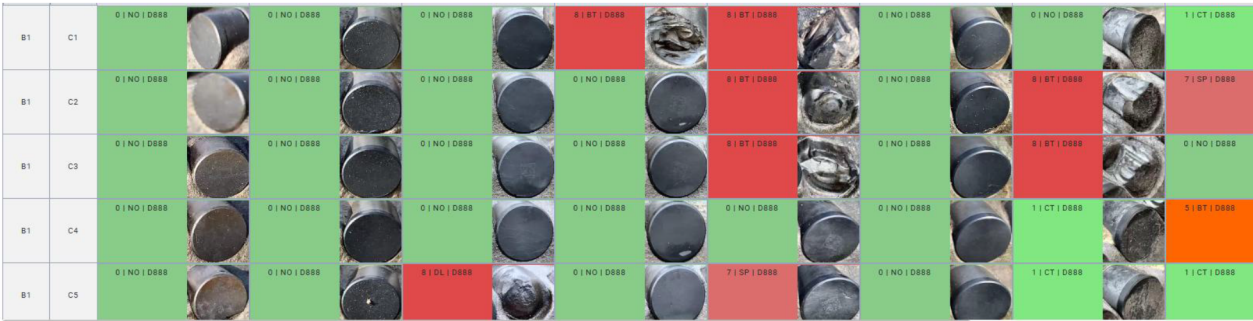


Figure 6—Dull grading cutter by cutter



Figure 6.1—Dull grading cutter by cutter

Blade	Cutter	Dull	Wear	Cracked	Erosion
B1 	C1	CT	T1	N	N
	C2	SP	T7	Y	N
	C3	NO	T0	N	N
	C4	BT	T5	Y	N
	C5	CT	T1	N	N
	C6	CT	T1	N	N
	C7	CT	T1	N	N
	C8	SP	T8	N	N
	C9	BT	T3	N	N
	C10	CT	T1	N	N
	C11	WT	T1	N	N
	C12	CT	T1	N	N
	C13	WT	T1	N	N

Figure 6.2—Cutter coverage and IADC dull and wear code

Comparing Dull Condition of the Design vs. Cutting Structure Work Rates and Normal Forces

To better understand the root causes of damage and validate analytical findings, the dull grading results were cross-referenced with simulated cutting structure work rates and normal forces under both static

and dynamic drilling conditions. This comparison was performed using proprietary bit design software capable of modeling cutter loading based on wellbore geometry, lithology, and real-time drilling data. The simulations illustrated how certain areas of the bit were exposed to higher normal or tangential forces. When overlaid with the damage zone heat maps, clear correlations were observed between zones of high loading and actual dull features. Further investigation linked cutter back rake angles, exposure levels, and position on the bit profile to the nature and severity of the damage, guiding the identification of potential design flaws or areas for improvement. (Fig. 7)

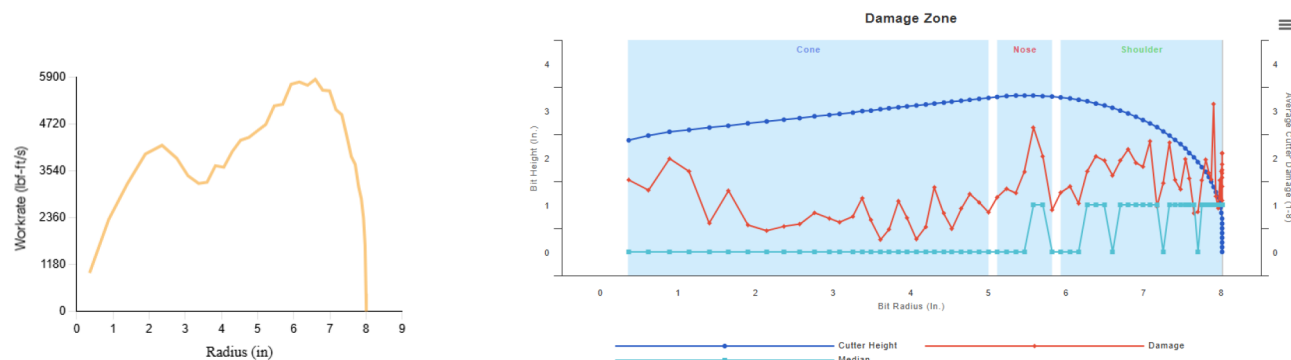


Figure 7—Workrate and Actual damage zone comparison

Designing of New Bit with Modified Profile and Cutting Structure

Based on detailed insights derived from the CA platform and mechanical loading simulations, a new drill bit design was proposed with the objective of mitigating cutter failure, improving load distribution, and enhancing overall durability. The analytics platform had revealed specific patterns of localized cutter wear and premature damage—particularly in the shoulder and nose areas—which served as critical inputs into the redesign process. These observations were further supported by mechanical stress simulations, which pinpointed zones of high contact pressure and uneven load transfer.

The redesign effort focused primarily on modifying both the bit profile geometry and the cutter layout to optimize performance under anticipated downhole conditions. A key priority was to redistribute mechanical loads more uniformly across the cutting structure to prevent overloading individual cutters. This was achieved by implementing several targeted changes in the design (Fig. 8):

- Adjustment of back rake angles on selected cutters to modify the aggressiveness of the cut and reduce impact loading. This also helped improve cutter entry angle and rock engagement in tougher formations.
- Optimization of cutter spacing along both the axial and radial axes. By recalibrating cutter placement, the design aimed to reduce cutter redundancy, avoid overlap in cutting paths, and promote smoother torque response.
- Reinforcement of shoulder areas, which had previously shown signs of chipping and spalling. This was achieved by placing higher-grade cutters and altering cutter orientation to better absorb lateral forces.
- Modification of the bit profile curvature, especially across the nose and shoulder transitions, to enable smoother weight transfer and reduce the risk of sudden mechanical shocks or impact-induced cutter failures.

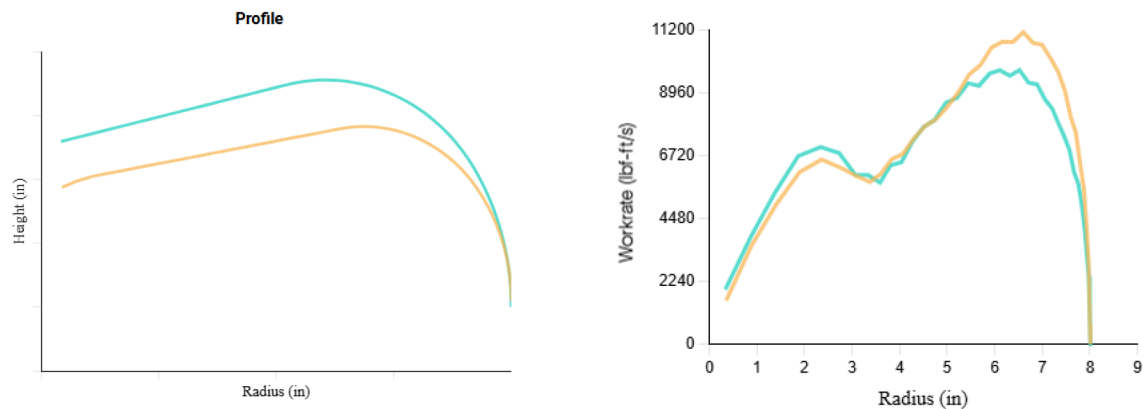


Figure 8—New design (purple line) profile change and workrate comparison vs. old baseline bit

The entire design process was supported by a digital engineering platform that enabled rapid prototyping, automated meshing, and finite element analysis (FEA) under simulated downhole conditions. This capability allowed multiple design iterations to be quickly generated and evaluated in terms of expected performance metrics such as torque response, bit aggressiveness index, cutter engagement depth, and stress distribution.

A significant advantage of this digital workflow was the speed and precision with which feedback loops could be closed. Results from each simulation cycle were fed back into the design loop, enabling designers to iteratively refine critical parameters without the time and cost associated with physical prototyping at each stage. The platform also integrated field data from previous runs, allowing real-world calibration of rock mechanics models and bit-rock interaction simulations.

Once the final geometry was selected and verified through simulation (including stress, load, and thermal analysis), a prototype bit was manufactured using advanced machining and brazing techniques. High-performance PDC cutters, selected based on prior durability assessments.

The prototype bit was then prepared for field testing in a similar lithological environment to the original run. Predictive models—based on drilling parameters such as WOB, RPM, flow rate, and formation type—indicated that the redesigned bit offered improved mechanical efficiency. This was evidenced by:

- More efficient energy transfer from the rotary system to the rock,
- Lower estimated cutter engagement forces, and
- Reduced cumulative cutter wear per unit depth drilled.
- Preliminary simulation results (see Fig. 8.1) showed a smoother load curve across the cutter array and a significant drop in localized peak stresses, especially in the shoulder region. These findings suggested that the new bit design would likely offer enhanced durability, slower wear progression, and better cutting efficiency—particularly in hard or interbedded formations where previous bits had experienced early-life cutter degradation.

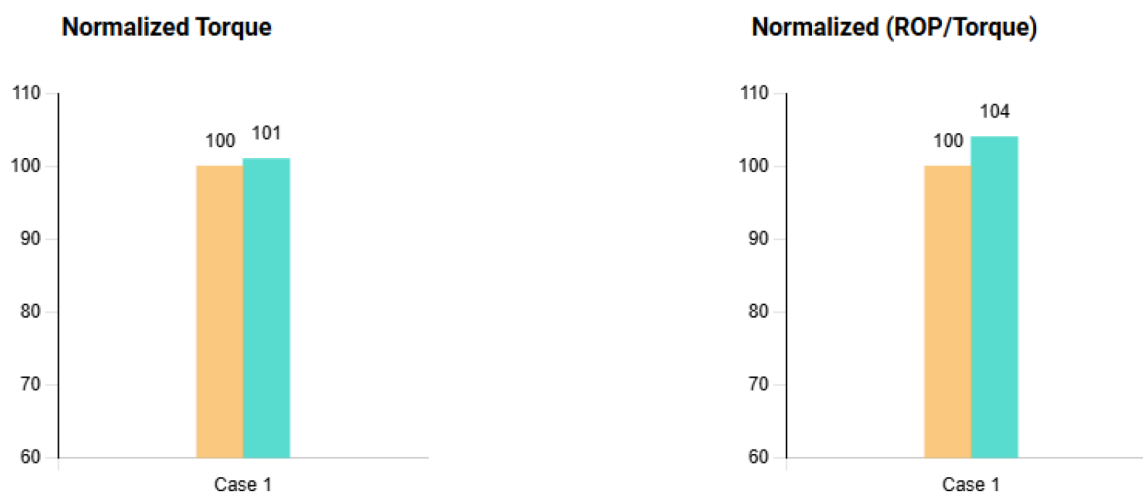


Figure 8.1—New design (purple bar) cutting structure efficiency comparison vs. old baseline bit

Moreover, the predictive dull grading, generated before the field trial, forecasted a more balanced wear pattern and fewer high-severity dull codes, particularly in the outer blades. This aligns with the design intent and validates the effectiveness of integrating digital tools and machine learning insights into the bit development process.

In conclusion, the redesign process showcased the power of combining high-resolution CA with advanced design and simulation tools. By systematically addressing the root causes of cutter failure observed in the previous bit runs, the new bit design is expected to deliver measurable improvements in both performance and operational longevity. Field deployment and post-run evaluation will provide the final verification of its effectiveness and may inform further refinements in the next design cycle.

Field Testing Bit and Performance Evaluation

The redesigned bit was deployed in the same 16-inch section application where previous bits had experienced frequent failures. During the run, comprehensive real-time data was collected, including ROP, torque, WOB, and vibration. Post-run evaluation showed that the new bit outperformed previous designs by delivering longer run length, improved ROP, and significantly reduced dull conditions upon retrieval. The post-run CA confirmed that damage was minimized, especially in previously high-failure zones. These results validated the effectiveness of the AI-driven analysis, and the design improvements derived from it.

This field test provided a full cycle proof of concept: from identifying failure patterns using photometric dull grading, analyzing them with AI and simulation tools, implementing design changes, and validating improvements through actual drilling performance. (Fig.9)

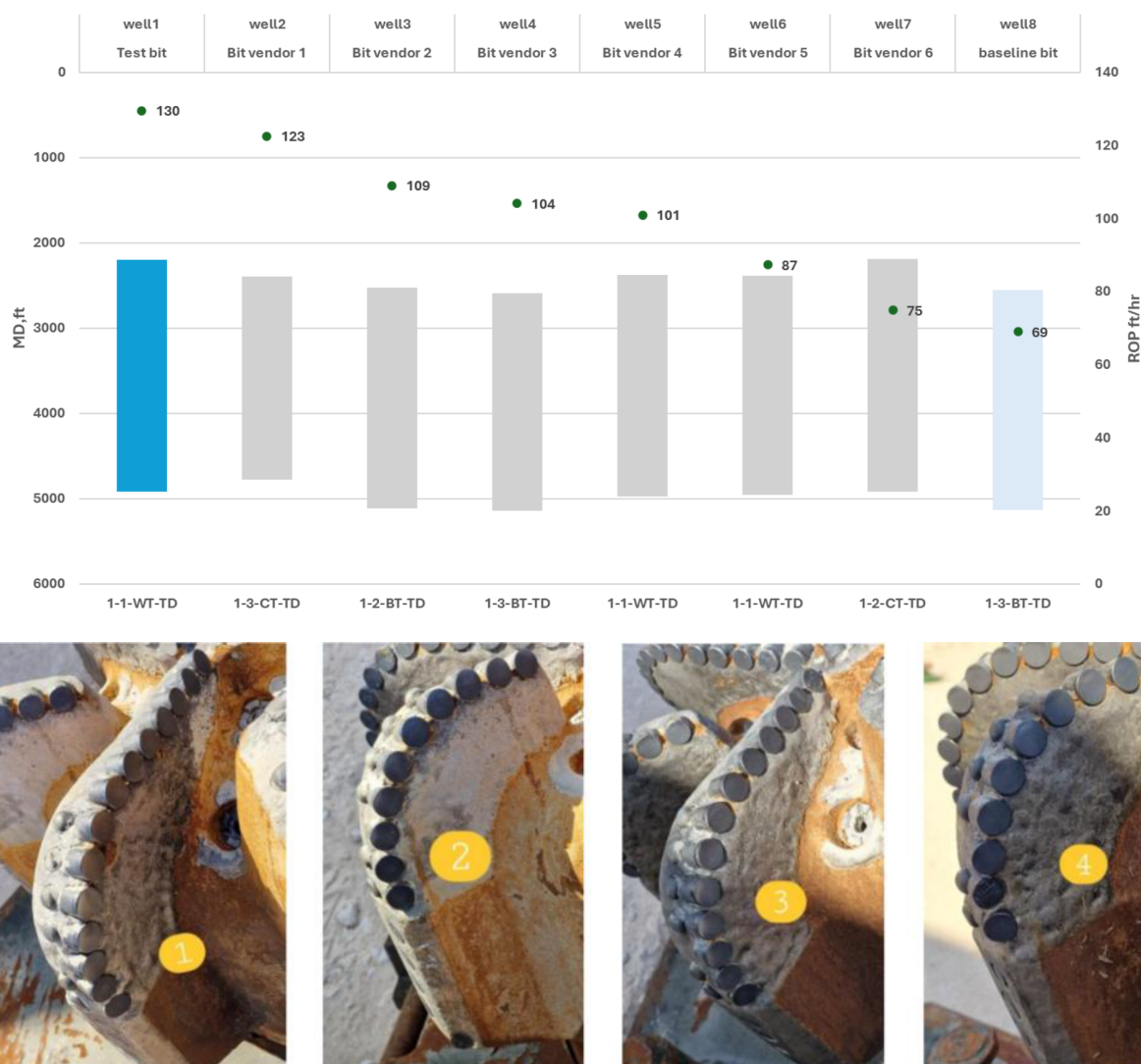


Figure 9—New design field test results and dull condition

Novel/Additive Information

CA technology is built on five core platforms: a mobile digital acquisition system, AI-driven learning for big data analytics and dull grading using advanced integrated view, mapping the dull grading codes to identify the weak points in bit cutting structure.

Enhanced bit cutting structure wear analysis can be done using the mobile app, available on Android and iOS, captures high-resolution (1080P) videos and images of bit dulls, which are linked to unique serial numbers and session IDs for comprehensive tracking throughout the bit's lifecycle. Additional metadata can also be stored with each scan.

ML algorithms, including Google Auto ML and Open CV, are employed for object detection and tracking to accurately grade cutter dullness. A custom algorithm identifies and orders primary and secondary cutters. The system's core capability is its integration with other databases containing drilling, run, and well data, providing a complete 360-degree view of the bit run for thorough dull forensics. Future integrations will include rock strength and in-bit sensory data.

Conclusion

The integration of photometric dull grading with AI-generated software has revolutionized the process of identifying the bit design cutting structure weakness. The AI software analyzes the dull grading data to identify the damage patterns and insights that inform the design of new bits.

The new bit designs created using this software have demonstrated significant performance and dull improvements, particularly in high-impact and challenging drilling application.

Field tests comparing the new design to baseline bits revealed remarkable enhancements in several key areas such as durability and ROP.

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